TATAWID: Traffic Assistive Technology for All Walking Individuals with Detection and Sensory Integration

by

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Chapter 1

INTRODUCTION

1.1 Background of the Study

Road traffic injuries and accidents are one of the leading causes of deaths to 15-29 year old Filipinos, including children [2]. In Metro Manila alone, roughly Php 76.6 million is lost annually due to road traffic incidents [1]. The Philippines has indeed implemented strong measures to reduce the statistics, involving the strict enforcement of driver's license and vehicle plate issuance [4]. However, it does not directly guarantee the safety of the crossing civilians, especially children and Person-With-Disabilities (PWDs).

Studies such as Crosswatch: a System for Providing Guidance to Visually Impaired Travelers at Traffic Intersections, extend support to the civilians, its functionality directly aimed to visually-impaired travelers [3]; Wearable systems utilizing AI, specifically designed to enhance the safety of visually impaired pedestrians at zebra crossings [5], are being researched and targeted internationally. Such measures, however, have yet to be approached in the Philippines.

Accordingly, the researchers devised an IoT-based approach to provide assistance to the general public, with PWDs and children as priority. Through sensor integration and movement detection, the researchers aim to design an efficient device that detects pedestrian and traffic movements.

As one of the most perturbing concerns in the Philippines, impacting the country physically, emotionally, and financially, steering the focus to the respective civilians presents a different framework to the current measures the country has to offer.

1.2 Statement of the Problem

Road safety is a priority that must be guaranteed to all road users. However, according to the World Health Organization, 1.3 million people die from road traffic crashes worldwide – which made it the 8th leading cause of death globally [2]. These crashes threatened the road safety of individuals, especially pedestrians – the most vulnerable road users. Contributing to the road safety plan or system for pedestrians could alleviate the number of road traffic crashes and provide safer roads to all road users.

1.3 Significance of the Study

The TATAWID project will be helpful to the following entities:

All Walking Individuals. This will improve pedestrian safety and mobility for all walkers, resulting in a safer environment.

Persons with Disabilities (PWDs). The project directly addresses pedestrian safety, independence, and quality of life by providing real-time alerts, and sensory feedback, significantly reducing the risk of accidents.

City Officials. It will offer insights and data to inform policy development and urban planning initiatives aimed at improving accessibility for all citizens.

Transportation Authorities. This will aid in developing policies and strategies that enhance pedestrian safety, and facilitate the implementation of assistive technologies in transportation systems.

Technology Developers and Engineers. The findings of this study will guide developers and engineers in creating more effective assistive devices and applications that cater to the needs of everyone.

Researchers. This will assist current researchers in developing more effective solutions related to sensors and assistive technologies designed to improve pedestrian safety and accessibility.

Future Researchers. Researchers looking to conduct related or comparable studies will benefit from this work, as it serves as an authoritative and reliable reference for related future studies.

1.4 Objectives of the Study

The study aims to develop a device that receives sensor data, which is then sent to a mobile application designed to assist the general public, especially people with disabilities (PWDs). Moreover, TATAWID intends to:

1. Produce vibrations and alerts upon receiving sensor data from device to application.
2. Detect traffic and vehicle movements using sensors.
3. Incorporate an armband design for the device.
4. Assess whether the user can proceed across the pedestrian crossing using the data received from the device, and notify accordingly.
5. Implement a user-friendly application able to cater PWDs, children, and the general public.
6. Map pedestrian crossing locations, referred to as "checkpoints", according to the user's usual route
7. Optimize the device to a suitable and accessible weight for children and PWDs.

1.5 Scope and Delimitation

The focus of the study is to develop a traffic assistive technology system designed primarily for all walking individuals, especially for people with disabilities (PWDs) and children. This study is limited to the following:

1. The wristband may not be able to implement its function to assess the ability to proceed across a pedestrian crossing in a heavy traffic flow because of little to no vehicle movement.

2. The wristband requires an optimized weight to achieve its full functionality.
3. The wristband, although chargeable, can only be used for a certain amount of time.
4. The wristband will only be able to detect movements within a distance limit of approximately 10 meters.
5. The mobile application is not its separate platform. It is only meant to receive sensor data and notify accordingly.
6. The mobile application will be developed exclusively for Android operating systems.

Chapter 2

REVIEW OF RELATED LITERATURE

2.1. THEORETICAL BACKGROUND

2.1.1. Road Traffic Safety

Road safety refers to the actions, measures, and precautions implemented to reduce the risk of accidents, injuries, and fatalities on roads and highways [2]. This includes obeying traffic signals, speed limits, and using safety equipment such as seat belts and helmets. Additional measures like pedestrian crossings, well-marked lanes, traffic lights, and speed bumps help control traffic flow and reduce risks for drivers, cyclists, pedestrians, and public transport users. The ultimate goal is to create a safer environment for all road users [1].

2.1.2. Assistive Technology for Pedestrian

Assistive technology for pedestrians refers to devices or systems that help individuals, especially those with visual impairments, navigate public spaces safely and independently. These technologies use sensors and feedback mechanisms to convey important environmental information, aiding in orientation and mobility [3]

2.1.3. Sensory Integration

Sensory integration is the process of receiving sensory information, organizing it, and using it to engage in daily activities [4]. Assistive products, including physical devices and digital solutions, are essential for over 2.5 billion people globally. By 2050, this number is expected to rise to 3.5 billion, but many individuals still lack access to these technologies [5]

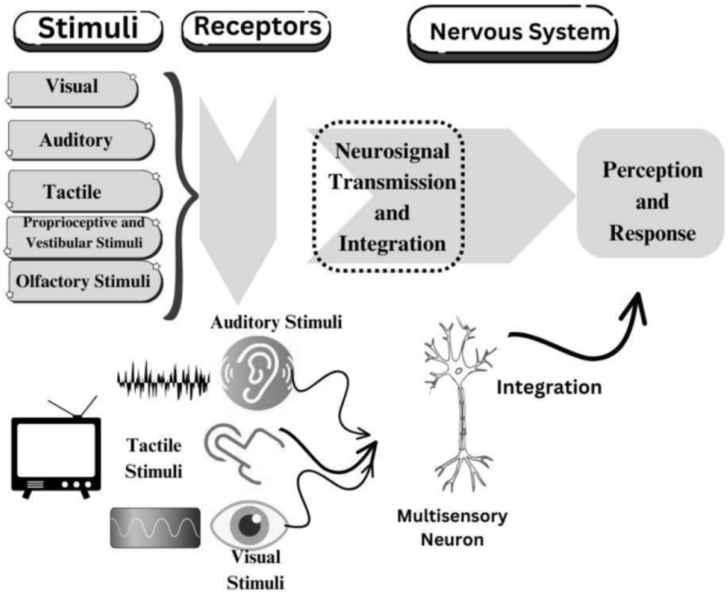

Fig. 2.1 Schematic representation of multi-sensory stimulation

2.1.4. Human Machine Interaction (HMI)

Human-machine interaction (HMI) involves the exchange of information and interaction between a human and a machine through a user interface. Recently, natural user interfaces, like gestures, have gained popularity because they enable users to control machines using intuitive and natural actions [6].

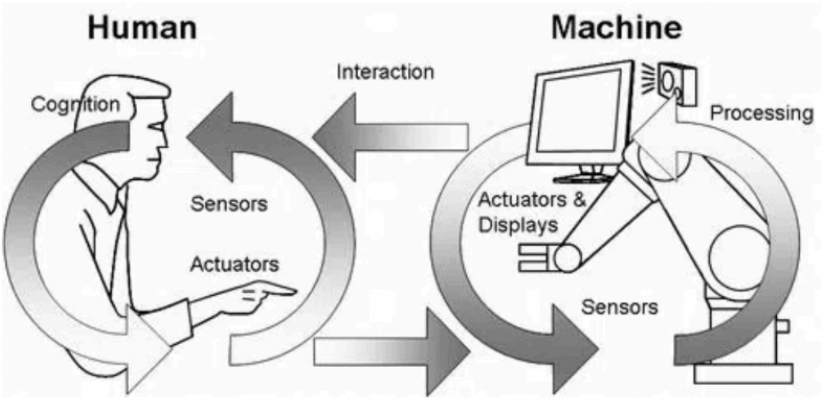

Fig. 2.2 Working of HMI – Human Machine Interface

2.1.5. Smart Wearables

Wearables are electronic devices integrated into items that can be worn on the body. These devices track real-time information by using motion sensors to record daily activities and sync the data with mobile devices or laptops[7]. Modern wearable technology encompasses a wide range of devices, including smartwatches, fitness trackers, VR headsets, and smart jewelry, each serving different purposes like health, fitness, or entertainment [8].

2.2 RELATED STUDIES

2.2.1. Assistive technology in smart cities: A case of street crossing for the visually-impaired

Visually-impaired individuals encounter significant challenges in daily tasks that are straightforward for the general population, such as crossing streets. Traditional solutions like Accessible Pedestrian Signals (APS), which rely on audio signals, have proven to be inadequate for many visually-impaired users. Given the growing accessibility of smartphones among this demographic, there exists an opportunity to provide enhanced solutions through mobile technology.

In this study in Taiwan, the researchers presented the Smart Accessible Pedestrian Signal System (SAPSS), a solution designed using ability-based principles to ensure accessibility. The system integrates hardware, software, and a mobile application to provide visually-impaired users with a safer and more confident street-crossing experience. Two field tests involving 19 visually-impaired participants were conducted to refine the system, and the positive feedback from users validated its usability. SAPSS has since been implemented in three cities in Taiwan,

demonstrating its potential to address limitations in traditional APS and contribute to the development of smart cities.

This study highlights the importance of integrating innovative assistive technologies to enhance mobility and confidence for visually-impaired individuals, emphasizing its implications for creating inclusive urban environments. [15]

2.2.2. Crosswatch: a System for Providing Guidance to Visually Impaired Travelers at Traffic Intersections

The "Crosswatch" project represents a significant advancement in smartphone-based systems designed to assist blind and visually impaired individuals at traffic intersections. The system builds on earlier developments by enhancing user alignment with crosswalks and interpreting walk signal statuses. Recent progress has focused on integrating computer vision with additional information sources, such as Geographic Information Systems (GIS) and sensor data, to achieve precise user location beyond GPS capabilities. This integration enables Crosswatch to provide a broader range of traffic intersection information, enhancing pedestrian safety and confidence.

Key functionalities of Crosswatch include precise location determination and a user-friendly interface tailored to visually impaired users. Preliminary testing of these functionalities demonstrated their feasibility and usability. Although these tests revealed areas for improvement, they underscore the system's potential. The study emphasizes the maturity of the technologies underpinning Crosswatch, suggesting its readiness for practical application in the near future.

This innovative approach addresses critical challenges faced by visually impaired travelers, offering a promising solution for improving their independence and mobility. The findings highlight the system's potential to become a practical tool for enhancing accessibility in urban environments. [3]

2.2.3. An Artificial Intelligence Edge Computing-Based Assistive System for Visually Impaired Pedestrian Safety at Zebra Crossings

In this study, a wearable assistive system based on artificial intelligence (AI) edge computing techniques is proposed to help visually impaired individuals safely navigate zebra crossings. The system consists of three components: smart sunglasses, a waist-mounted intelligent device, and an intelligent walking cane. Real-time zebra crossing recognition is enabled by a deep learning algorithm integrated into the system.

Visually impaired users are required to wear the smart sunglasses and waist-mounted device while holding the intelligent walking cane. Upon approaching a zebra crossing, the system provides immediate feedback regarding the current situation at the crossing and the status of the traffic light signal. Experimental evaluations demonstrate the system’s high efficacy, achieving up to 90% accuracy in real-time zebra crossing recognition.

This study highlights the potential of AI-powered wearable technology to enhance the safety and independence of visually impaired pedestrians, offering a practical solution to navigating urban environments. [5]

2.2.4. A Wearable Assistive Device for Blind Pedestrians Using Real-Time Object Detection and Tactile Presentation

Traffic safety is a common concern especially for visually-impaired people. In this study, utilizing object detection methods and novel tactile display made from shape-memory alloy (SMA) actuators, the researchers presented a wearable device which employs various tactile sensations to notify the user of the position of the obstacles detected in real-time.

Considering the level of comfort and quality of the resulting device, the researchers aimed to develop the device as compact as possible, therefore conducting the programming and computation to a mobile device. With the device, the researchers aim to support the reduction of risks and hazards for the prioritized users' day-to-day living.

For the object detection device, a slimming compression method was introduced to reduce 90% of the repetitive neural network structures which were restricted by the memory limitations.. This resulted in a quicker inference speed when implemented on a mobile device. Several experiments were conducted, and exhibited positive results. Navigation for subjects was pursued without collisions in most situations, wherein the subject was able to determine the position of a motionless obstacle with about 96% accuracy while using the wearable device, which can be inferred to as the device effectively executing its roles and features [14].

Chapter 3

METHODOLOGY

This chapter discusses the operations, techniques, and methods to be utilized in the study.

3.1 System's Model

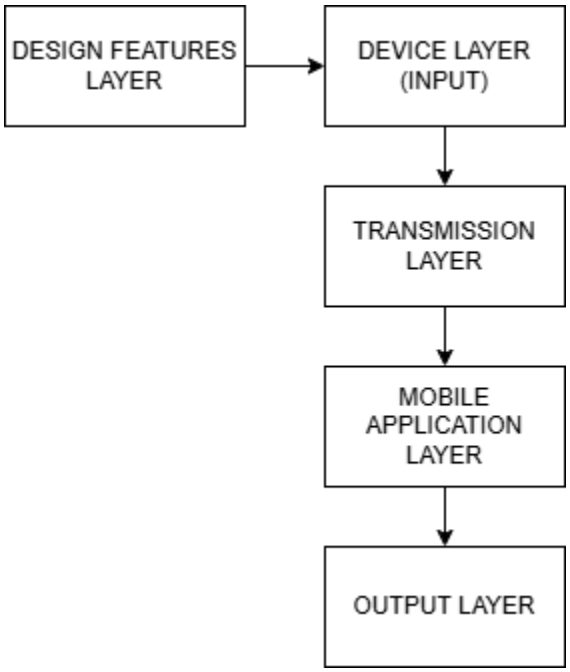

Figure 2.3 Block Diagram of the System Model

As presented in Figure 2.3, the block diagram of the system model, the system begins with a data input from the hardware device, which is then processed and sent to the mobile application through the transmission layer. The user is notified through the notifications and vibrations as produced in the output layer.

3.1.1 Design Features Layer

This layer integrates an armband design for the hardware segment of the device, housing the sensors and microcontrollers, ensuring that the device remains lightweight and suitable for PWDs and children.

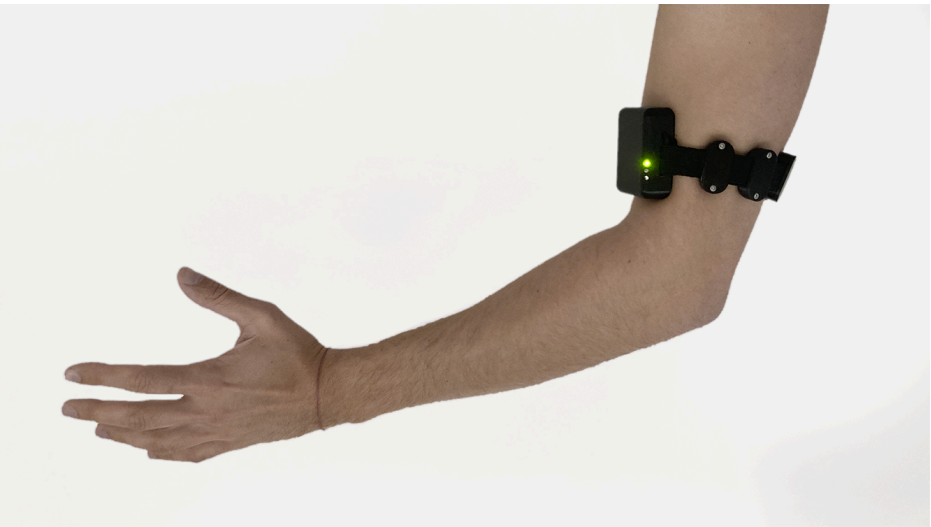

Figure 2.4 Armband Design

3.1.2 Device Layer (Input)

The device layer serves as the data acquisition and processing layer of the system, which contains a sensors module which detects traffic and vehicle movement, and a dedicated unit including a camera module, infrared and ultrasonic sensors, and LIDAR (Light Detection and Ranging) with image, heat signature, size, distance, and proximity recognition for human/vehicle recognition; a separate microcontroller with machine learning capabilities for data processing; and a data fusion unit for better classification accuracy. Pre-trained classification models are stored in the microcontrollers' memory for initialization between human and vehicle characteristics.

3.1.3 Transmission Layer

The transmission layer includes a wireless communication module, centering on sending processed sensor data to the mobile application.

3.1.4 Mobile Application Layer

Once sensor data is transmitted from the device layer, it is analyzed for pedestrian crossing assessment. A notification system is integrated, providing vibrations and alerts to allow the user to proceed with their route, in consideration to the following key factors: traffic speed and proximity, signal and crosswalk data, environmental conditions such as nearby engine noises or honks, weather conditions, and human readiness. The analysis module evaluates the time gap between the vehicles and the users to ensure a proper crossing based on the user's walking speed. The user's common route is mapped to determine pedestrian crossings and checkpoints, and create a customized profile for a quicker and more user-friendly interface.

3.1.5 Output Layer

The output layer displays the notifications and crossing status to the user in the mobile application. The mobile device where the application is installed will also be providing vibrations based on the results of the data processed by the analysis module of the mobile application.

3.2 Program Flowchart

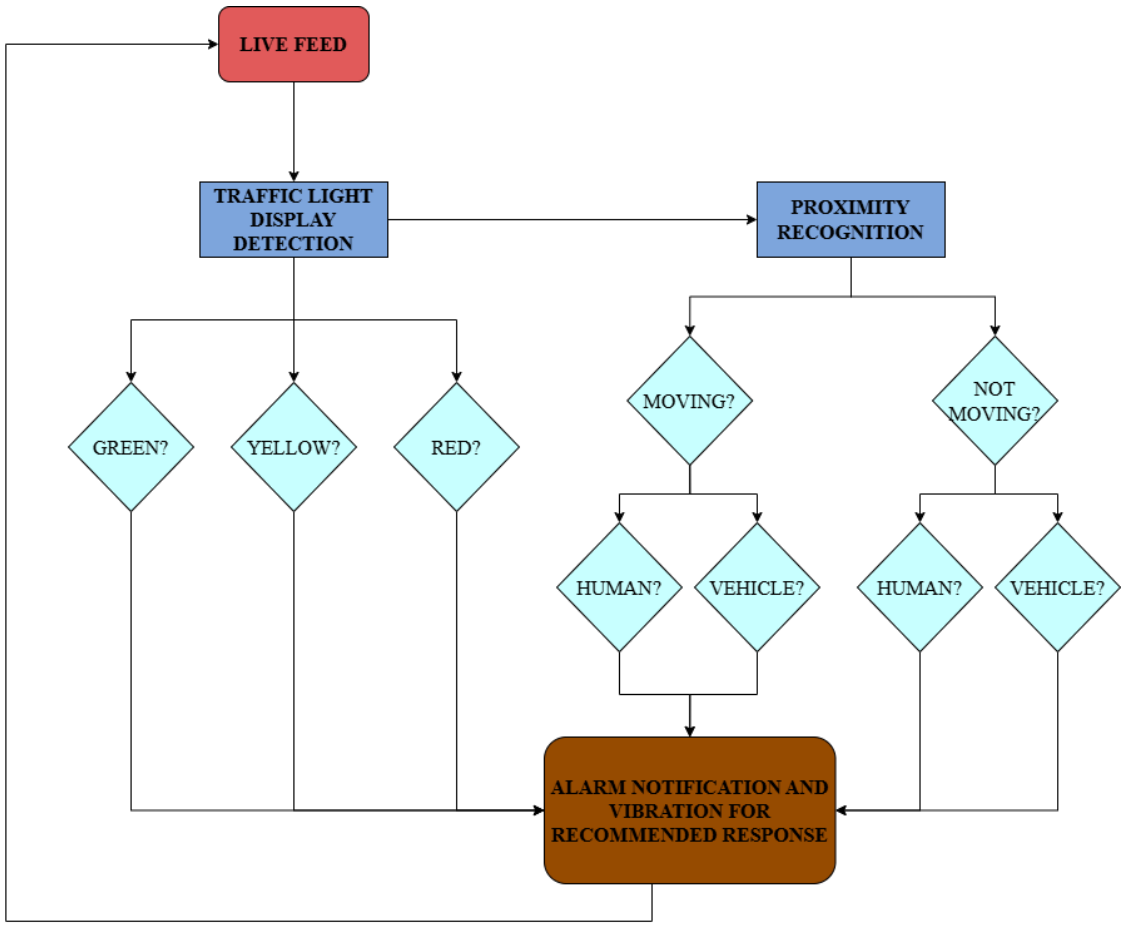

Figure 2.5 Alarm Notification and Vibration Flowchart

Figure 2.5 is the program flowchart that demonstrates how TATAWID integrated multiple sensory inputs to provide real-time, actionable feedback for pedestrians, ensuring their safety in crossing roads. The system uses sensory integration (live feed and proximity detection) to create an intelligent and user-centric traffic assistive tool.

3.3 Physical Setup

The physical setup of this study involves a wearable armband and a mobile application, both wirelessly connected. The armband is equipped with sensors to monitor pedestrian crossing

conditions, while the mobile app processes the data and provides real-time notifications to ensure safe crossings. This wireless design ensures convenience and ease of use, especially for individuals with disabilities and children, while enhancing pedestrian safety without hindering mobility.

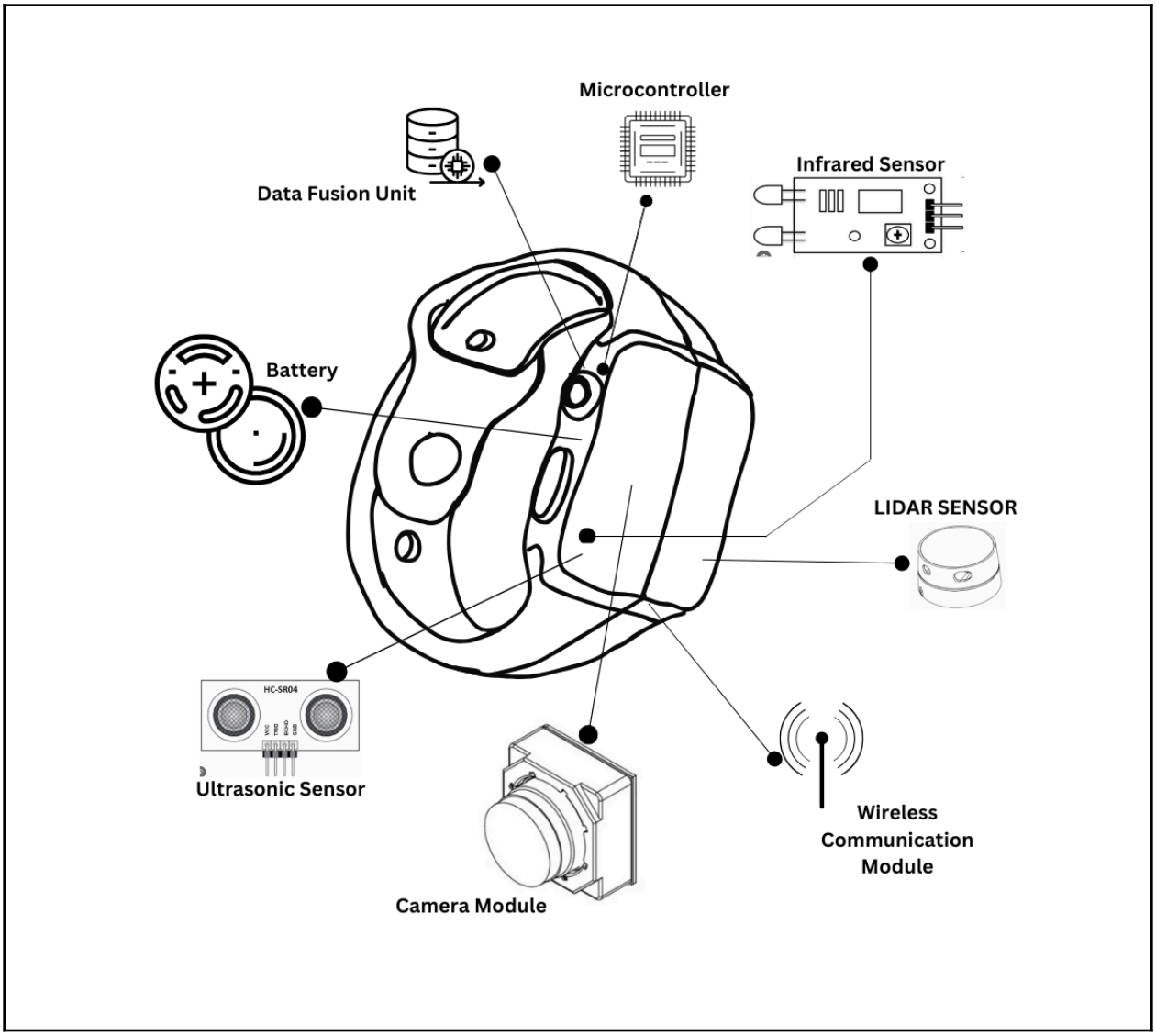

Figure 2.6 Physical Setup of the Armband

Figure 2.5 shows the physical connection of the main device used in this study. This device, designed to resemble a watch or armband, integrates multiple sensors and components to facilitate its functions. The device includes an Infrared Sensor positioned on the side of the watch

face to detect nearby objects, and an Ultrasonic Sensor placed similarly to measure distances and detect obstacles. The LIDAR (Light Detection and Ranging) Sensor, also located on the side or front, scans the surroundings to identify human and vehicle movements. The Camera Module is installed on the front face of the watch, capturing images and providing visual data for further analysis. Inside the watch casing, a Microcontroller Unit processes data from all sensors, and a Data Fusion Unit enhances classification accuracy by integrating data from multiple sources. The device also contains a Wireless Communication Module, which transmits the processed data to a mobile application, and a Power Supply, such as a battery, ensuring the device remains operational.

These components are arranged to maintain a lightweight, comfortable design, making it suitable for people with disabilities (PWDs) and children. The armband ensures the device is practical for everyday use while effectively monitoring and assessing pedestrian crossing conditions.

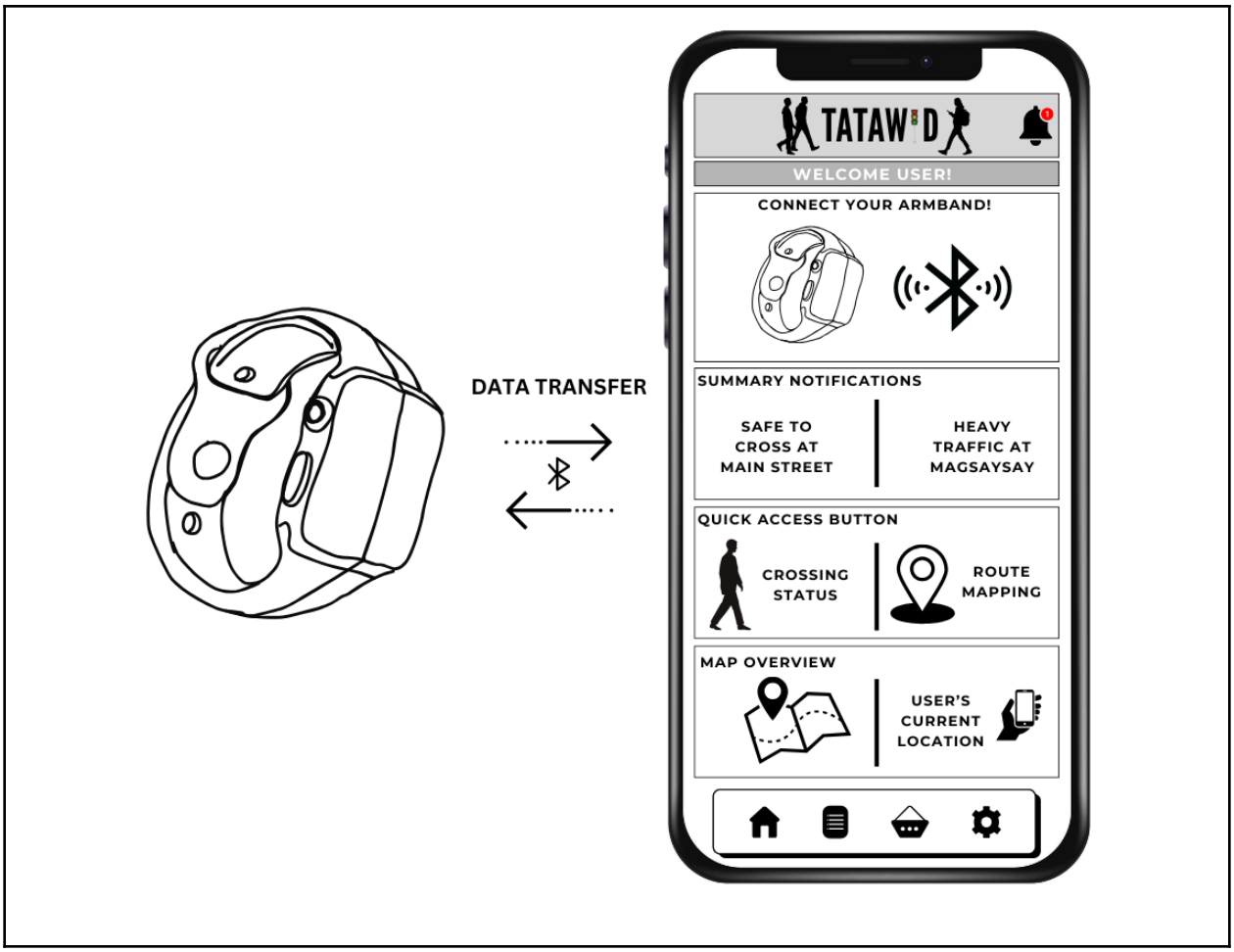

Figure 2.7 Physical Setup of Armband and Mobile App Interface

Figure 2.6 shows the physical setup of the armband and mobile app interface, illustrating how the wearable device, equipped with sensors and microcontrollers, captures real-time data related to pedestrian crossing conditions. The armband transmits this data wirelessly to the mobile application, which processes and analyzes the information to provide users with notifications and alerts. This system is designed to ensure safe and efficient pedestrian navigation by assessing traffic speed, proximity, and environmental factors. The figure demonstrates the seamless integration of the hardware and software components to enhance user safety and awareness.

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